

2016 US Presidential Elections: Information Warfare Patient Zero

Behavioural Fingerprinting, Narrative Strategy, and Temporal Coordination in the Internet Research Agency's 2016 Twitter Campaign

Working Paper

Abstract

This paper presents a multi-method computational analysis of 203,461 tweets produced by 454 confirmed Internet Research Agency (IRA) accounts during the 2016 United States presidential election cycle. Drawing on the NBC News dataset, we pursue four research questions: (1) what behavioural signatures distinguish the IRA account network; (2) what narrative clusters structured the IRA's messaging strategy; (3) how did IRA sentiment toward Trump and Clinton differ; and (4) what operational patterns characterise the network's architecture. We find that IRA accounts exhibit a median dormancy period of 781 days between creation and activation - more than four times the 177-day benchmark reported by Symantec - indicating systematic multi-year pre-registration infrastructure. Latent Dirichlet Allocation (LDA) identifies eight topic clusters, the two largest being culturally-oriented content targeting Black Twitter (15.6%) and Tea Party/conservative messaging (14.1%), consistent with the IRA's documented dual-track strategy of both mobilising conservative audiences and suppressing Black voter enthusiasm. VADER sentiment analysis reveals a statistically significant asymmetry in how the IRA framed the two candidates: tweets mentioning Trump cluster near neutrality (mean = -0.009), while Clinton-mentioning tweets are systematically negative (mean = -0.076 ; Mann-Whitney U, $p = 9.14 \times 10^{-52}$, $r = -0.084$). K-means clustering (silhouette = 0.746) identifies four account typologies: Cultural Infiltrators (86%), Dormant Nodes (10%), and a small core of Amplifier Bots and Human Operators (2% each). Lorenz-curve analysis of retweet amplification reveals extreme inequality (Gini = 0.985), with a handful of accounts capturing the entirety of external engagement. Z-score-based spike detection identifies 18 anomalous days in 2016, including the highest single-day volume on 6 October 2016 - the eve of the Access Hollywood tape and WikiLeaks Podesta email release. *Together, these findings characterise the IRA's operation as a pre-planned, architecturally stratified, event-reactive infrastructure optimised for narrative saturation rather than direct persuasion.*

Keywords: Internet Research Agency, computational propaganda, influence operations, Twitter, LDA, sentiment analysis, behavioural fingerprinting, disinformation

1. Introduction

The disclosure of the Internet Research Agency's (IRA) Twitter operations by the United States Senate Select Committee on Intelligence in 2018 provided the most systematically documented case of state-sponsored computational propaganda in democratic politics. The IRA - a Russian

government-linked entity based in St. Petersburg - deployed thousands of accounts across multiple platforms from approximately 2013 onward, with activity accelerating markedly during the 2016 U.S. presidential election cycle. The operation's core mechanics - persona creation, narrative injection, amplification, and cross-platform coordination - have become a reference model for the study of influence operations in adversarial information environments.

Despite substantial scholarly attention to the IRA's campaigns, several dimensions of the operation's architecture remain underspecified in the existing literature. Most studies operate at scale - millions of tweets, thousands of accounts - and optimise for breadth over precision. The behavioural fingerprints of individual account typologies, the internal amplification inequality of the network, and the event-driven temporal structure of IRA posting have received less systematic treatment than content-level analyses. This paper addresses these gaps through a multi-method computational analysis of a confirmed IRA subset: 203,461 tweets from 454 accounts released by NBC News, spanning July 2014 to September 2017.

We pursue four research questions:

- RQ1 (Behavioural Fingerprinting): What account-level signatures - including dormancy period, posting velocity, follower/friend asymmetry, and timezone configuration - characterise the IRA network?
- RQ2 (Narrative Strategy): What content clusters structured the IRA's messaging, and how did their prevalence shift across the electoral timeline?
- RQ3 (Candidate Targeting): How did IRA sentiment toward Trump and Clinton differ, and is this difference statistically robust?
- RQ4 (Operational Architecture): What clustering, amplification, and temporal patterns characterise the IRA's network structure?

Our contributions are threefold. First, we apply K-means account typology clustering with silhouette-score validation to identify four functionally distinct account archetypes within the IRA network. Second, we conduct Lorenz-curve and Gini coefficient analysis of retweet amplification, revealing extreme concentration ($Gini = 0.985$) that challenges assumptions of distributed influence. Third, we implement z-score-based spike detection calibrated to the electoral calendar, identifying anomalous posting surges aligned with specific high-salience events. Together these methods move beyond descriptive content analysis toward a structural characterisation of the operation's architecture.

2. Background and Related Work

2.1 The IRA's 2016 Operation

The IRA's Twitter presence during the 2016 election has been documented across multiple authoritative sources. The Senate Intelligence Committee's bipartisan report established that the IRA operated accounts collectively accumulating 30 million retweets and reaching approximately 126 million Facebook users. The Mueller Report's Volume I detailed the IRA's structural organisation - its 'translator department,' its unit specialisations by target audience - while stopping short of attributing electoral causation. Twitter itself disclosed 2,752 IRA-linked accounts to the Senate in 2017, a dataset substantially larger than the one analysed here.

The IRA's strategic logic has been characterised in two partially competing frameworks. The 'pro-Trump' thesis (Linville & Warren, 2020) holds that the operation primarily served to boost Republican turnout and suppress Democratic enthusiasm. The 'discord sowing' thesis (Bail et al., 2019; Golovchenko et al., 2020) emphasises the IRA's investment in both ends of the ideological spectrum, targeting liberal and conservative audiences with content designed to exacerbate divisions rather than simply win converts. Our findings suggest these framings are complementary rather than exclusive: the operation deployed cultural infiltration at scale while maintaining a smaller core of ideologically sharpened amplifier accounts.

2.2 Prior Computational Analyses

Linville and Warren (2018) produced the most widely cited content-level taxonomy of IRA accounts, categorising them into five functional types: Right Trolls, Left Trolls, News Feeds, Hashtag Gamers, and Fearmongers. Their analysis of approximately 3 million tweets found Right Trolls to be the most engaged and prolific. Badawy et al. (2018) extended this work through network analysis of 13 million election-related tweets, finding that conservative users who retweeted IRA content generated approximately eight times more engagement than their liberal counterparts. Bail et al. (2019), using a nationally representative panel with matched Twitter data, found no evidence that IRA interaction substantially shifted political attitudes - though they noted the study's temporal limitations and the selection effects of treating interaction as the unit of analysis.

Golovchenko et al. (2020) conducted a cross-platform analysis of hyperlinks in 108,781 IRA tweets, finding that conservative-facing accounts were consistently more active and that YouTube was the second-most linked destination. Bovet and Makse (2019), analysing 171 million tweets from 47 million users, found that IRA accounts reached approximately 16 million Twitter users during the election campaign. Collectively, these studies establish the scale, ideological asymmetry, and amplification reach of the IRA operation, but leave underexplored the internal behavioural architecture of the account network itself - the gap this paper addresses.

3. Data and Methods

3.1 Dataset

The analysis draws on the Russian Troll Tweets dataset released by NBC News and hosted on Kaggle (Kaggle dataset ID: vikasg/russian-troll-tweets), comprising two files: tweets.csv (203,461 tweets after cleaning; originally identified in the NBC reporting on IRA-linked accounts) and users.csv (454 account records). The dataset spans 14 July 2014 to 26 September 2017, with peak activity in the October–November 2016 window. This is a confirmed, validated subset - all accounts were publicly attributed to the IRA by NBC's reporting - and is therefore more suitable for characterising IRA-specific behaviour than larger datasets that mix confirmed and probabilistically attributed accounts.

After removing null-text rows, the working corpus consists of 203,461 tweets: 163,810 original posts (80.5%) and 39,651 retweets (19.5%). 88,785 tweets (43.6%) contain at least one hashtag; 29,693 (14.6%) contain at least one URL. The 454 accounts comprise 273 English-language (60.1%) and 90 Russian-language (19.8%) accounts, with 257 (56.6%) configured with US timezones and 104 (22.9%) using Russian or Eastern European timezones - a discrepancy with reported locations that is itself analytically significant.

Metric	Value
Total tweets (cleaned)	203,461
Unique accounts	454
Date range	14 Jul 2014 - 26 Sep 2017
Original posts	163,810 (80.5%)
Retweets	39,651 (19.5%)

Tweets with hashtags	88,785 (43.6%)
Tweets with URLs	29,693 (14.6%)
English-language accounts	273 (60.1%)
Russian-language accounts	90 (19.8%)
US-timezone accounts	257 (56.6%)
Russian/E. European TZ accounts	104 (22.9%)
Median account dormancy	781 days
Mean account dormancy	626 days

Table 1. Dataset summary statistics.

3.2 Behavioural Feature Engineering

For each account we computed five behavioural features from the merged tweet and user records. Dormancy period is defined as the number of days between the account's creation date (from `users.created_at`) and the timestamp of its earliest tweet in the dataset. Posting velocity is total tweet count divided by the number of active days between first and last tweet. The follower/friend ratio (`followers_count ÷ friends_count`) proxies for the account's asymmetric reach - high ratios indicate broadcast-oriented accounts with large audiences relative to their following. Retweet share measures the proportion of an account's tweets that are retweets of other content. Average retweets received proxies for external engagement and amplification.

3.3 Topic Modelling

We applied Latent Dirichlet Allocation (LDA) to the English-language tweet corpus (35,912 tweets from 273 accounts). Text preprocessing involved lowercasing, URL and mention removal, hashtag word retention (`#` stripped, term kept), punctuation and digit removal, stopword filtering against a combined English stopwords list augmented with Twitter-specific noise terms (`rt`, `amp`, `co`) and the candidate names themselves (`trump`, `clinton`, `hillary`, `donald`) - the latter exclusion ensuring that topic clusters surface thematic content rather than mere mention patterns. Basic suffix-stripping lemmatisation was applied. We used `CountVectorizer` with a vocabulary of 6,000 features, minimum document frequency of 5, maximum document frequency of 0.85, and bigram support (n-gram range 1–2). LDA was fitted with 8 components, online learning with batch size 4,096, and 25 iterations. The choice of $k=8$ reflects the IRA's documented multi-audience strategy; we found it produced the most interpretable topic structure after inspection of top words.

3.4 Sentiment Analysis

We applied VADER (Valence Aware Dictionary and sEntiment Reasoner) to all 203,461 tweets, extracting compound scores bounded on $[-1, 1]$. VADER was selected over lexicon-based alternatives (e.g., AFINN) for its calibration to social media text, including sensitivity to capitalisation, punctuation repetition, and emoji - features characteristic of politicised Twitter content. For candidate-specific analyses, tweets were filtered by regex patterns for `\btrump\b` and `\b(hillary|clinton)\b` respectively. To test whether observed sentiment differences are statistically significant, we applied the Mann-Whitney U test on the 2016 subsample, selecting this non-parametric alternative to Student's t-test given the bimodal, non-normal structure of the VADER compound distribution (visible in Figure 7). Effect size is reported as rank-biserial correlation r .

3.5 Account Clustering

We applied K-means clustering ($k=4$, $n_init=20$, $max_iter=500$) to the five behavioural features described in 3.2, after StandardScaler normalisation and 99th-percentile outlier capping. Principal Component Analysis ($k=2$) was applied for visualisation. The silhouette score (0.746) indicates strong cluster separation. Semantic labels were assigned post-hoc by inspecting cluster centroids ordered by posting velocity. Cluster validity is supported both by the silhouette score and by convergence with the content-based typologies documented in prior IRA research.

3.6 Lorenz Curve and Gini Coefficient

To characterise inequality in network amplification, we computed Lorenz curves and Gini coefficients for three distributions: average retweets received per account, follower count, and total tweet volume. The Gini coefficient ($G = 1 - 2\int \text{Lorenz}$) was computed via the trapezoidal rule on the empirical cumulative distribution. $G = 0$ represents perfect equality; $G = 1$ represents maximal concentration. This methodology, standard in inequality economics, has not to our knowledge been applied systematically to IRA account-level amplification distributions.

3.7 Spike Detection

Daily tweet volume was computed for the 2016 calendar year. A 7-day centred rolling mean and standard deviation were used to compute daily z-scores ($z = (x - \mu) / \sigma$). Days with $|z| > 2.0$ were flagged as statistically anomalous. This threshold corresponds approximately to the 95th percentile under a normal distribution and was calibrated to the observed variance structure of the dataset.

Detected spikes were cross-referenced against the electoral event calendar to test for event-reactivity.

4. Results

4.1 RQ1 - Behavioural Fingerprinting

The most striking finding in the behavioural analysis concerns dormancy period. IRA accounts in this dataset exhibit a median dormancy of 781 days between account creation and first observed tweet - more than two years - with a mean of 626 days (SD = 458 days; Figure 2, Panel A). This substantially exceeds the 177-day benchmark reported in Symantec's analysis of IRA accounts (Symantec, 2019), which was itself already considered evidence of pre-planned infrastructure. The distribution is right-skewed with a maximum of 2,458 days, indicating that some accounts were registered as early as 2009 and remained dormant until 2016. This pattern is inconsistent with ad hoc account creation and strongly suggests the maintenance of a pre-registered account reserve available for activation at operationally determined moments.

Posting velocity exhibits high variance across the network (Figure 2, Panel B). The majority of accounts post at a moderate pace, consistent with human-level production, while a tail of accounts post at velocities suggesting automation or near-continuous operation. The follower/friend ratio distribution (Figure 2, Panel C) is predominantly below 1.0, meaning most accounts follow more accounts than follow them back - consistent with an influence-acquisition strategy common to inauthentic accounts in the early phase of persona construction. A subset of accounts show elevated ratios, indicating they have successfully cultivated audiences.

The timezone analysis reveals a meaningful discrepancy between claimed location and account configuration. 104 accounts (22.9%) are configured with Russian or Eastern European timezones, despite the majority claiming US-based locations (Figure 3). This aligns with documented IRA operational slippage reported in the Senate Intelligence Committee's analysis. Among Russian-language accounts, Eastern European timezones are nearly universal, while English-language accounts predominantly use US timezones - suggesting that the US-facing English-language track received more attention to location cover.

4.2 RQ2 - Narrative Strategy

LDA topic modelling of the English-language corpus (35,912 tweets) identifies eight topic clusters spanning the ideological spectrum of the IRA's documented targeting strategy (Figure 4). The two largest clusters by corpus share are Tea Party/Conservative messaging (15.6%) and Black Culture/Targeting (14.2%), followed by Police/Anti-Media content (14.1%). These proportions are substantively significant: the Black Culture cluster, characterised by music, entertainment, and cultural reference terms (top words: black, show, friend, man, listen, music, best, today), represents a well-documented IRA tactic of building credibility within Black Twitter communities before introducing politically divisive content.

Topic	Label	% Corpus	Top Words
1	Black Culture / Targeting	14.2%	black, show, friend, man, listen, music, today, best
2	Anti-Obama / Email Scandal	11.9%	obama, president, state, campaign, email, election, fbi
3	White House / Political	11.1%	white, live, house, hate, american, bill, care, call
4	Tea Party / Conservative	15.6%	black, thing, tcot, gop, obama, american, big, free, work
5	Police / Anti-Media	14.1%	obama, kill, police, media, country, need, election
6	Tea Party / Anti-Islam	10.0%	teapartynew, conservative, usa, muslim, america, theteaparty
7	Religious / Ambiguous	12.6%	word, alway, right, god, really, take, tweet, thank, game
8	Pro-Trump / MAGA	10.6%	maga, vote, neverhillary, love, america, trump Pence, great

Table 2. LDA topic model summary ($k=8$, English-language corpus, $n=35,912$ tweets). % Corpus = share of tweets assigned this dominant topic.

The Pro-Trump / MAGA cluster, despite its high salience in public discourse about the IRA, accounts for only 10.6% of the English-language corpus - a smaller share than Black Culture, Tea Party/Conservative, or Police/Anti-Media content. This is consistent with the argument that the IRA's operation was primarily a discord-amplification strategy rather than a direct pro-Trump campaign, even as its effect at the margins may have been asymmetrically beneficial to the Republican candidate.

Topic prevalence over time (Figure 5) reveals meaningful temporal dynamics. Black Culture content is consistently prominent from mid-2015 onward, declining relatively as Pro-Trump/MAGA and Anti-Clinton content rise through the summer and autumn of 2016. Police/Anti-Media content spikes in September 2016, consistent with the coverage period of the Charlotte, North Carolina protests and other police-related events. The Anti-Islam/Tea Party

cluster shows elevated prevalence in 2015, declining into 2016 as the operation's electoral focus sharpened.

4.3 RQ3 - Candidate Targeting and Sentiment Asymmetry

VADER sentiment analysis of 2016 tweets reveals a statistically robust asymmetry in how the IRA framed the two candidates. Tweets mentioning Trump ($n = 24,382$) produce a mean compound score of -0.009 (essentially neutral), while tweets mentioning Clinton ($n = 18,111$) produce a mean of -0.076 ($p = 9.14 \times 10^{-52}$, Mann-Whitney U; effect size $r = -0.084$). Figure 6 plots the 7-day rolling mean sentiment time series for both candidates over September–November 2016, with electoral events annotated.

Metric	Trump	Clinton
N (tweets, 2016)	24,382	18,111
Mean VADER score	-0.009	-0.076
Median VADER score	0.000	-0.034
Mann-Whitney U	239,406,962	-
p-value	9.14×10^{-52}	-
Effect size (r)	-0.084 (small)	-

Table 3. VADER sentiment comparison: Trump-mentioning vs. Clinton-mentioning tweets (2016 subsample). Mann-Whitney U selected for non-normal compound score distribution.

The effect size ($r = -0.084$) is small by conventional standards but the practical significance is non-trivial: across tens of thousands of tweets produced over months of high-attention election coverage, the systematic negative framing of Clinton and near-neutral treatment of Trump constitutes a consistent directional bias in the IRA's content production. The KDE and violin plots (Figure 7) show that the Trump distribution is more concentrated around zero, while the Clinton distribution has a heavier left tail - the IRA generated more strongly negative content about Clinton than about any other subject.

4.4 RQ4 - Operational Architecture

K-means clustering ($k=4$; silhouette = 0.746) identifies four functionally distinct account typologies (Figure 11). The dominant typology - accounting for 389 of 454 accounts (85.7%) - is Cultural Infiltrators: accounts characterised by moderate posting velocity, English-language content, and cultural/entertainment framing consistent with the Black Culture targeting cluster

identified in the topic model. The second-largest group is Dormant Nodes (n=45, 9.9%), accounts with minimal activity likely representing the pre-registered account reserve described above. A small core of Amplifier Bots (n=10, 2.2%) and Human Operators (n=10, 2.2%) together constitute the operation's active hard core, distinguished by higher posting velocity, higher retweet share, and elevated external engagement respectively.

Typology	N	%	Key Features
Cultural Infiltrators	389	85.7%	Moderate velocity, cultural/entertainment content, English-language, audience-building profile
Dormant Nodes	45	9.9%	Very low activity, minimal engagement - likely pre-registered reserve accounts
Amplifier Bots	10	2.2%	High velocity, high RT share, low organic engagement - mechanistic content amplification
Human Operators	10	2.2%	Lower velocity, high follower ratio, higher external engagement - likely human-operated cores

Table 4. Account typology clusters (*K*-means, *k*=4; silhouette score = 0.746).

Lorenz-curve analysis of amplification inequality reveals extreme concentration (Figure 12). The Gini coefficient for average retweets received is 0.985 - approaching the theoretical maximum - indicating that essentially all external engagement with IRA content was concentrated in a tiny fraction of accounts. The Gini for follower count is 0.773 and for total tweet volume is 0.965. This pattern is consistent with a hub-and-spoke amplification architecture: a small number of high-visibility accounts (likely the Human Operators cluster) generate the content that achieves external reach, while the Cultural Infiltrator majority builds background credibility without individual amplification.

Distribution	Gini Coefficient	Interpretation
External retweets received	0.985	Near-total concentration in < 5% of accounts
Follower count	0.773	Substantially unequal - most accounts have small audiences
Total tweets	0.965	High concentration - a minority of accounts produce most content

Table 5. Gini coefficients for three amplification-related distributions across 454 IRA accounts.

Z-score spike detection identifies 18 anomalous posting days in 2016 ($|z| > 2.0$; Figure 13). The single highest-volume day in the entire dataset is 6 October 2016, with 3,861 tweets - the eve of both the Access Hollywood tape release and the WikiLeaks Podesta email disclosure. The second

and third highest days are 17 and 19 September 2016, corresponding to the period immediately following the Chelsea Manning bombing in New York and the Dahir Adan stabbing in Minnesota - events that the IRA's Anti-Islam and Police clusters were structurally oriented to amplify. Election Day itself (8 November 2016) registers as the third-highest single day at 2,867 tweets. This pattern of reactive spikes aligned with high-salience external events suggests that the IRA operated in an opportunistic-reactive mode during the election cycle, deploying pre-existing narrative infrastructure in response to unfolding news rather than attempting to set the agenda from below.

5. Discussion

The findings presented here converge on a portrait of the IRA's Twitter operation that is more structurally sophisticated than descriptive content analyses alone would suggest. Three interpretive threads are worth developing.

First, the dormancy finding reframes the operation's timeline. A median pre-registration period of 781 days means the IRA was building its Twitter infrastructure in the 2013–2014 period, well before the 2016 primary campaign had defined the candidates. This is not an operation that emerged in response to Trump's candidacy or Clinton's nomination - it was a standing capability that was directed toward the election. This has implications for how influence operations are theorised: the IRA's operation was less a targeted information campaign than a pre-positioned influence infrastructure that could be activated and oriented opportunistically. The available evidence suggests the election was the activation event, not the rationale for the operation's existence.

Second, the dominance of the Cultural Infiltrators cluster (85.7% of accounts) and the Black Culture/Targeting topic cluster (14.2% of corpus) speaks to a strategic logic that is under-appreciated in coverage focused on partisan political content. The IRA invested heavily in building authentic-seeming cultural presences - accounts that posted music recommendations, entertainment commentary, and social commentary before deploying more explicitly political content. This audience-cultivation strategy is recognisable from commercial social media marketing and is considerably more sophisticated than simple propaganda injection. The

implication is that detection methodologies that focus exclusively on political content will systematically undercount IRA-type operations.

Third, the amplification Gini of 0.985 tells a structural story that complements the typology analysis. If a handful of accounts generate essentially all external amplification, then the operation's reach is not a function of the number of accounts but of the performance of that small core. The Cultural Infiltrator majority may function less as direct influence agents and more as social proof - establishing the apparent grassroots character of the narrative clusters being driven by the Human Operator and Amplifier Bot cores. This hub-and-spoke interpretation reframes the question of what 'scale' means in the context of influence operations.

Several limitations warrant acknowledgment. This dataset is a confirmed but non-exhaustive subset of IRA accounts - the FiveThirtyEight dataset, released by Twitter, comprises over 2,700 accounts and 9 million tweets. Findings here are specific to the NBC-released subset and may not generalise to the full operation. LDA topic quality is constrained by the relatively small English-language corpus (35,912 tweets); a larger corpus would likely produce more granular and stable topics. The account clustering silhouette score (0.746), while strong, is partly an artifact of the large Cultural Infiltrators cluster dominating the feature space. Finally, the absence of control group data - non-IRA accounts from the same period - precludes direct testing of the behavioural fingerprint's discriminating power.

6. Conclusion

This paper has applied a multi-method computational analysis to 203,461 tweets from 454 confirmed IRA accounts, producing findings across four research questions. We have shown that IRA accounts exhibit a median dormancy of 781 days - more than four times the Symantec benchmark - consistent with a pre-positioned infrastructure rather than a purpose-built electoral operation. LDA identifies eight topic clusters, the largest being cultural content targeting Black Twitter communities, a finding that emphasises the IRA's audience-cultivation investment over its explicitly political content. Sentiment analysis reveals a statistically robust asymmetry in the framing of the two candidates ($p = 9.14 \times 10^{-52}$), with Clinton-mentioning content systematically more negative. Clustering analysis identifies four account typologies dominated by Cultural Infiltrators (85.7%), with a small high-intensity core of Amplifier Bots and Human Operators.

Amplification inequality is extreme (Gini = 0.985), suggesting hub-and-spoke network architecture. Event-driven spike detection identifies 18 anomalous days in 2016 aligned with high-salience electoral events, indicating reactive rather than agenda-setting operational behaviour.

These findings have practical implications for platform governance and influence operation detection. Dormancy-based features and amplification Gini coefficients are strong candidates for real-time detection metrics. The predominance of culturally-oriented content in the IRA's account mix suggests that detection frameworks must extend beyond political content classifiers. The reactive temporal pattern implies that high-salience news events constitute a window of elevated operation risk that platforms might prioritise for enhanced monitoring.

Future work should extend this analysis to the full FiveThirtyEight IRA dataset, apply transformer-based topic modelling (BERTopic) to improve topic coherence, and develop a multi-period comparative framework that tracks the IRA's strategy evolution across distinct electoral cycles. The intersection of behavioural fingerprinting and network analysis - specifically, testing whether the Gini structure and account typology clusters hold across differently attributed IRA datasets - represents a promising avenue for methodological generalisation.

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